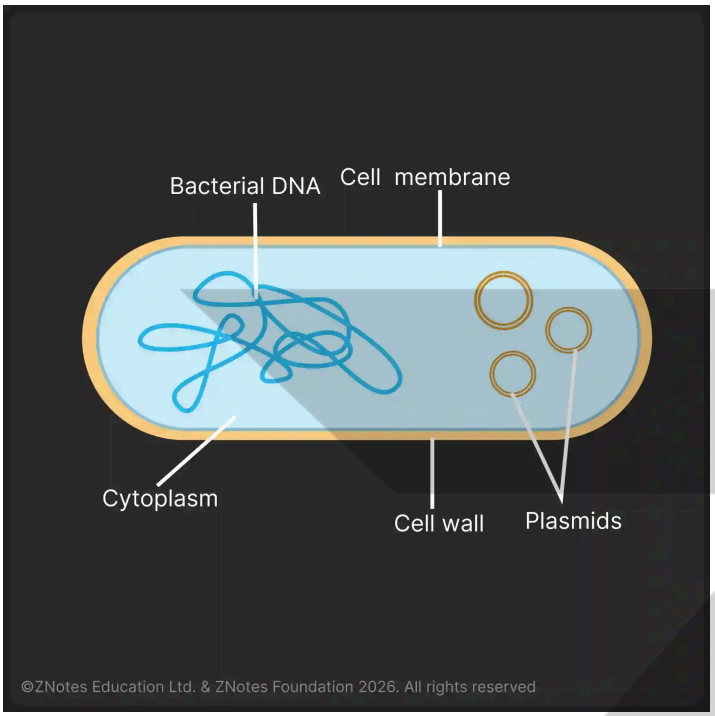
1. Biotechnology and Genetic Modification

1.1. Introduction



Bacteria are useful due to their rapid reproduction rate and ability to make complex molecules.

Why are bacteria useful in biotechnology and genetic modification?

1. There are a few ethical concerns over their manipulation and growth
2. The presence of plasmids
3. Producing complex molecules
4. Replicates rapidly

1.2. Biotechnology

Biofuel

- Use plants to make sugars, which yeast then breaks down to make ethanol.
- This process also uses anaerobic respiration.

Bread-Making

- Flour, sugar, water and salt are mixed with yeast to make the dough.
- Amylase breaks down some starch to make maltose and glucose. This is used by yeast in respiration.
- The dough is kept warm and moist (28°C). Yeast ferments sugar, making carbon dioxide, which creates bubbles, so bread rises.
- Cooking (at 180°C) – kills yeast, evaporates alcohol and hardens the outer surface.

Use of Enzymes in Biotechnology

Pectinase:

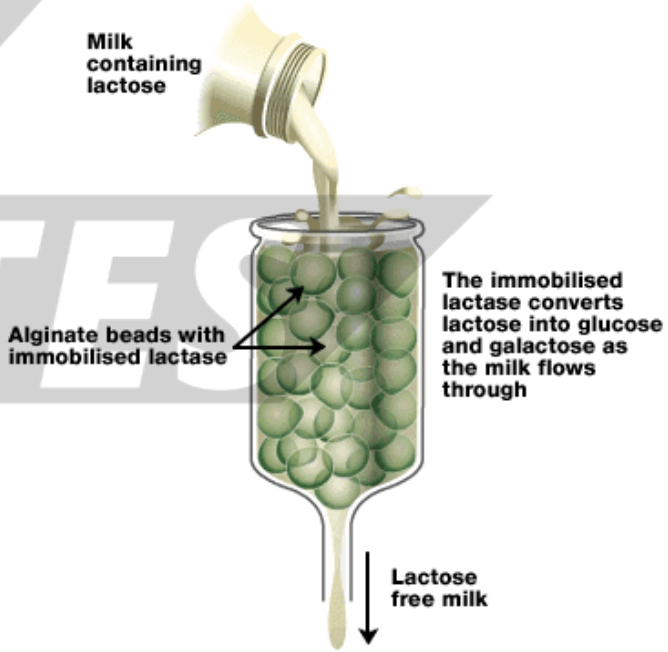
- Fruit juices are extracted using pectinase (breaks down pectin)
- Pectin helps plant walls stick together
- If pectin is broken down, it's easier to squeeze juice from the fruit
- Extraction of juice from fruit, making juice clear, not cloudy

Biological Washing powders:

- Biological washing powders and liquids contain enzymes that help remove the stain
- The enzymes are coated with a special wax that melts in the wash, releasing the enzyme
- Once the stains have been broken down, they are easier for detergents to remove
- Proteases: break down proteins in stains, e.g., grass, blood
- Lipases: break down stains containing fats and oil
- Amylases: break down carbohydrate-based stains, such as starch
- Cellulases: break down cellulose fibres

Lactase:

- The enzyme that breaks down lactose (the sugar found in milk), people can stop making lactase naturally and, therefore, can't digest lactose.



Lactose-free milk production

- Lactase is made from yeast
- Lactase bound to the surface of alginate beads
- Milk passed down beads
- Lactose is broken down into glucose and galactose
- Immobilized enzymes are reused

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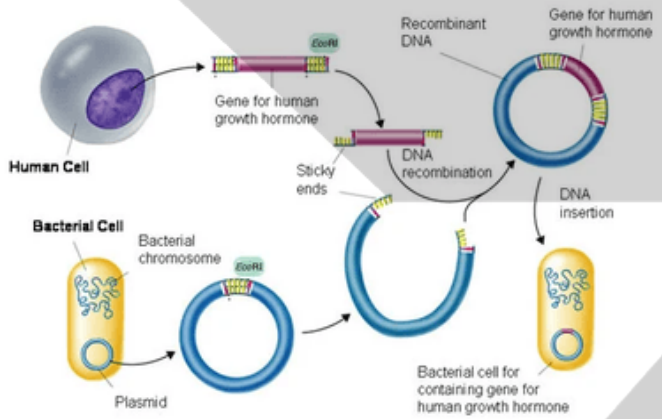
1.3. Genetic Modification

Genetic Modification: changing the genetic material of an organism by removing, altering, or inserting individual genes

- **Examples of genetic modification:**
  - the insertion of human genes into bacteria to produce human insulin
  - the insertion of genes into crop plants to confer resistance to herbicides
  - the insertion of genes into crop plants to confer resistance to insect pests
  - the insertion of genes into crop plants to provide additional vitamins

Human Insulin in Bacteria



- Isolation of the DNA making up a human gene using restriction enzymes, forming sticky ends.
- Cutting of bacterial plasmid DNA with the same restriction enzymes, forming complementary sticky ends.
- Insertion of human DNA into bacterial plasmid DNA using DNA ligase to form a recombinant plasmid – insertion of the plasmid into bacteria.
- Replication of bacteria containing recombinant plasmids, which make human protein as they express the gene

1.4. Fermenters

- **Penicillin:** an antibiotic produced by a fungus called **Penicillium**.
- They require proper temperature, pH, oxygen, nutrient supply and waste products.
- The stainless steel fermentation vessel contains a medium containing sugars and ammonium salts.
- Penicillium is added to produce penicillin. They use sugar for respiration and ammonium salts to make protein and nucleic acids
- The fermentation vessel consists of '**PAWS**'
  - **Probes** monitor temperature and pH
  - **Air** provides oxygen for aerobic respiration in fungus
  - A **water-cooled jacket** removes heat to maintain a temperature of 24°C.
  - **Stirrer** keeps the microorganism suspended (allowing access to nutrients and oxygen) while maintaining an even temperature.
- It is filtered to remove fungus and then can be crystallized to make capsules.

Mycoprotein

- The fungus Fusarium is cultured (grown) on an industrial scale in fermenters.
- These fermenters are large vats that can be kept at the optimum pH and temperature to grow.
- The fungus is grown in aerobic conditions and provided with glucose syrup as a food source.
- The fungus grows and multiplies within the fermenter
- It is then harvested and purified to produce mycoprotein
- Mycoprotein is a protein-rich food suitable for vegetarians, eg. It is used in QuornTM products (a brand of meat substitute products).

1.5. Genetically Modified Crops

Advantages	Disadvantages
Uniform in shape – easy to transport/appeal to consumers	Natural species may die
Growing season shorter	Decrease biodiversity/genetic diversity
Drought resistant – less water	Led to the development of superweeds – stronger than GM
Higher yields	No one knows the long-term effects on humans
Solve global hunger	Expensive seeds

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